

First Partial Exam (theory)

A. ALGEBRAIC STRUCTURES

1. Relations:

A triple $r = (A, B, R)$, where A, B are sets and

$$R \subseteq A \times B = \{(a, b) \mid a \in A, b \in B\},$$

is called a *(binary) relation*.

The set A is called the *domain*, the set B is called the *codomain* and the set R is called the *graph* of the relation r .

If $A = B$, then the relation r is called *homogeneous*.

Recall that a relation $r = (A, B, R)$ is called *homogeneous* if $A = B$.

Definition

A homogeneous relation $r = (A, A, R)$ on A is called:

- (1) *reflexive* (r) if: $\forall x \in A, x r x$.
- (2) *transitive* (t) if: $x, y, z \in A, x r y \text{ and } y r z \implies x r z$.
- (3) *symmetric* (s) if: $x, y \in A, x r y \implies y r x$.

A homogeneous relation $r = (A, A, R)$ is called an *equivalence relation* if r has the properties (r), (t) and (s).

2. Operations:

Definition

By an *operation* (or *composition law*) on a set A we understand a function

$$\varphi : A \times A \rightarrow A.$$

Usually, we denote operations by symbols like $\cdot, +, *$, so that $\varphi(x, y)$ is denoted by $x \cdot y, x + y, x * y, \forall (x, y) \in A \times A$. We denote by $(A, \cdot)$ the fact that " $\cdot$ " is an operation on a set A .

Definition

Let " $\cdot$ " be an operation on an arbitrary set A .

- (1) *Associative law*: $(x \cdot y) \cdot z = x \cdot (y \cdot z), \quad \forall x, y, z \in A$.
- (2) *Commutative law*: $x \cdot y = y \cdot x, \quad \forall x, y \in A$.
- (3) *Identity law*: $\exists e \in A$ such that $\forall a \in A, a \cdot e = e \cdot a = a$. In this case, e is called an *identity element*.
- (4) *Inverse law*: $\forall a \in A, \exists a' \in A$ such that $a \cdot a' = a' \cdot a = e$, where e is the identity element. In this case, a' is called an *inverse element* for a .

Lemma

Let " $\cdot$ " be an operation on a set A .

- (i) If there exists an identity element in A , then it is unique.
- (ii) Assume further that the operation " $\cdot$ " is associative and has identity element e and let $a \in A$. If an inverse element for a does exist, then it is unique.

3. Groups:

Definition

Let “ $\cdot$ ” be an operation on a set A . Then $(A, \cdot)$ is called a:

- (1) *semigroup* if the associative law holds.
- (2) *monoid* if it is a semigroup with identity element.
- (3) *group* if it is a monoid in which every element has an inverse.

If the operation is commutative as well, then the structure is called *commutative*. A commutative group is also called an *abelian group* (after the name of N. H. Abel).

Remark

We denote by 1 the identity element of a group $(G, \cdot)$ and by x^{-1} the inverse of an element $x \in G$. In case of an additive group $(G, +)$, the identity element is denoted by 0, while the inverse of an element $x \in G$ is called the *symmetric* of x and is denoted by $-x$.

- power in groups:

Let $(G, \cdot)$ be a semigroup, let $x \in G$ and let $n \in \mathbb{N}^*$. Then we may use the associative law and define

$$x^n = \underbrace{x \cdot x \cdot \dots \cdot x}_{n \text{ times}}.$$

If $(G, \cdot)$ is a monoid, then we may also define $x^0 = 1$.

If $(G, \cdot)$ is a group, then we may also define $x^{-n} = (x^{-1})^n$.

If the operation is “+”, then x^n becomes nx .

- examples:

(a) The operation “ $-$ ” defined on $\mathbb{Z}$ is not associative.

(b) $(\mathbb{N}^*, +)$ is a semigroup, but not a monoid.

(c) $(\mathbb{N}, +)$, $(\mathbb{N}, \cdot)$, $(\mathbb{Z}, \cdot)$, $(\mathbb{Q}, \cdot)$, $(\mathbb{R}, \cdot)$, $(\mathbb{C}, \cdot)$ are monoids, but not groups.

(d) $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, $(\mathbb{C}, +)$, $(\mathbb{Q}^*, \cdot)$, $(\mathbb{R}^*, \cdot)$ and $(\mathbb{C}^*, \cdot)$ are groups.

(e) Let X be a non-empty set. By a *word* on X of length n we understand a string of n elements from X for some $n \in \mathbb{N}$. The word of length 0 is called the *void word* and is denoted by e . On the set X^* of words on X consider the operation “ $\cdot$ ” given by concatenation. Then $(X^*, \cdot)$ is a monoid with identity element e , called the *free monoid* on the set X .

(f) Let $\{e\}$ be a single element set and let “ $\cdot$ ” be the only operation on $\{e\}$, defined by $e \cdot e = e$. Then $(\{e\}, \cdot)$ is an abelian group, called the *trivial group*.

(g) Let $n \in \mathbb{N}$, $n \geq 2$. Then $(\mathbb{Z}_n, +)$ is an abelian group, called the *group of residue classes modulo n* . The addition is defined by

$$\widehat{x} + \widehat{y} = \widehat{x + y}, \quad \forall \widehat{x}, \widehat{y} \in \mathbb{Z}_n.$$

(h) Let $n \in \mathbb{N}$ with $n \geq 2$. Denote by $M_{m,n}(\mathbb{R})$ the set of $m \times n$ -matrices with entries in $\mathbb{R}$ and by $M_n(\mathbb{R})$ the set of $n \times n$ -matrices with entries in $\mathbb{R}$. Then $(M_{m,n}(\mathbb{R}), +)$ is an abelian group and $(M_n(\mathbb{R}), \cdot)$ is a monoid. Denote by $GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$ the set of invertible $n \times n$ -matrices with real entries. Then $(GL_n(\mathbb{R}), \cdot)$ is a group, called the *general linear group of rank n* .

(i) Let M be a set and let $S_M = \{f : M \rightarrow M \mid f \text{ is bijective}\}$. Then $(S_M, \circ)$ is a group, called the *symmetric group of M* . The identity element is the identity map 1_M and the inverse of an element f (which is a bijection) is the inverse function f^{-1} . If $|M| = n$, then S_M is denoted by S_n , and the group $(S_n, \circ)$ is in fact the *permutation group of n elements*.

(j) Let $K = \{e, a, b, c\}$ and define an operation " $\cdot$ " on K by the following table:

$\cdot$	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	e	a
c	c	b	a	e

Then $(K, \cdot)$ is an abelian group, called *Klein's group*. It may be viewed as the group of geometric transformations of a rectangle:

- e is the identical transformation,
- a is the symmetry with respect to the horizontal symmetry axis,
- b is the symmetry with respect to the vertical symmetry axis,
- c is the symmetry with respect to the center of the circumscribed circle.

The product $x \cdot y$ is defined by performing first y and then x .

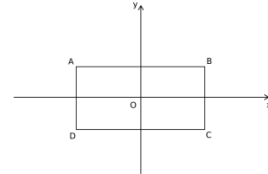


Figure: Klein's group.

4. Rings:

Definition

Let R be a set. A structure with two operations $(R, +, \cdot)$ is called a:

(1) *ring* if $(R, +)$ is an abelian group, $(R, \cdot)$ is a semigroup and the *distributive laws* hold:

$$x \cdot (y + z) = x \cdot y + x \cdot z, \quad \forall x, y, z \in R,$$

$$(y + z) \cdot x = y \cdot x + z \cdot x, \quad \forall x, y, z \in R.$$

(2) *unitary ring* if $(R, +, \cdot)$ is a ring and there is an identity element with respect to " $\cdot$ ".

(3) *division ring* (or *skew field*) if $(R, +)$ is an abelian group, $(R^*, \cdot)$ is a group and the distributive laws hold.

(4) *field* if it is a commutative division ring.

The ring $(R, +, \cdot)$ is called *commutative* if " $\cdot$ " is commutative.

- power in rings:

Let $(R, +, \cdot)$ be a ring, let $x \in R$ and let $n \in \mathbb{N}^*$. Then we define

$$n \cdot x = \underbrace{x + x + \cdots + x}_{n \text{ times}},$$

$$0 \cdot x = 0,$$

$$(-n) \cdot x = -n \cdot x,$$

$$x^n = \underbrace{x \cdot x \cdot \cdots \cdot x}_{n \text{ times}}.$$

If R is a unitary ring, then we may also consider $x^0 = 1$.

If R is a division ring, then we may also define $x^{-n} = (x^{-1})^n$.

- examples:

- (a) $(\mathbb{Z}, +, \cdot)$ is a unitary ring, but not a field.
 (b) $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$ and $(\mathbb{C}, +, \cdot)$ are fields.
 (c) Let $\{e\}$ be a single element set and let both “+” and “ $\cdot$ ” be the only operation on $\{e\}$, defined by $e + e = e$ and $e \cdot e = e$. Then $(\{e\}, +, \cdot)$ is a commutative unitary ring, called the *trivial ring*.
 (d) Let $n \in \mathbb{N}$, $n \geq 2$. Then $(\mathbb{Z}_n, +, \cdot)$ is a commutative unitary ring, called the *ring of residue classes modulo n*. The addition and the multiplication are defined by

$$\widehat{x} + \widehat{y} = \widehat{x + y}, \quad \widehat{x} \cdot \widehat{y} = \widehat{x \cdot y}, \quad \forall \widehat{x}, \widehat{y} \in \mathbb{Z}_n.$$

Note that $(\mathbb{Z}_n, +, \cdot)$ is a field if and only if n is prime.

- (e) Let $(R, +, \cdot)$ be a commutative unitary ring. Then $(R[X], +, \cdot)$ is a commutative unitary ring, called the *polynomial ring over R in*

the indeterminate X , where the operations are the usual addition and multiplication of polynomials.

- (f) Let $n \in \mathbb{N}$, $n \geq 2$ and let $(R, +, \cdot)$ be a ring. Then $(M_n(R), +, \cdot)$ is a ring, called the *ring of matrices $n \times n$ with entries in R*, where the operations are the usual addition and multiplication of matrices.

- (g) Let M be a non-empty set and let $(R, +, \cdot)$ be a ring. Define on the set

$$R^M = \{f \mid f : M \rightarrow R\}$$

two operations by: $\forall f, g \in R^M$, we have $f + g : M \rightarrow R$, $f \cdot g : M \rightarrow R$, where

$$(f + g)(x) = f(x) + g(x), \quad \forall x \in M,$$

$$(f \cdot g)(x) = f(x) \cdot g(x), \quad \forall x \in M.$$

5. Subgroups:

Definition

Let $(G, \cdot)$ be a group and let $H \subseteq G$. Then H is called a *subgroup* of G if:

- (i) H is a stable subset of $(G, \cdot)$.
 (ii) $(H, \cdot)$ is a group.

- characterization of subgroups:

Let $(G, \cdot)$ be a group and let $H \subseteq G$. Then

$$H \leq G \Leftrightarrow \begin{cases} H \neq \emptyset \ (1 \in H) \\ \forall x, y \in H, x \cdot y \in H \\ \forall x \in H, x^{-1} \in H. \end{cases} \Leftrightarrow \begin{cases} H \neq \emptyset \ (1 \in H) \\ \forall x, y \in H, x \cdot y^{-1} \in H. \end{cases}$$

In case of an additive group $(G, +)$, the conditions become:

- $\forall x, y \in H, x + y \in H$.
- $\forall x \in H, -x \in H$.
- $\forall x, y \in H, x - y \in H$.

- examples:

- (a) Every non-trivial group $(G, \cdot)$ has two subgroups, namely $\{1\}$ and G , called the *trivial subgroups*.

- (b) $\mathbb{Z}$ is a subgroup of $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$, $\mathbb{Q}$ is a subgroup of $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$, $\mathbb{R}$ is a subgroup of $(\mathbb{C}, +)$.

- (c) The set

$$n\mathbb{Z} = \{nx \mid x \in \mathbb{Z}\}$$

is a subgroup of $(\mathbb{Z}, +)$ for every $n \in \mathbb{N}$.

- (d) The set

$$H = \{z \in \mathbb{C} \mid |z| = 1\}$$

is a subgroup of the group $(\mathbb{C}^*, \cdot)$, called the *circle group*. But it is not a subgroup of the group $(\mathbb{C}, +)$.

- (e) The set

$$U_n = \{z \in \mathbb{C} \mid z^n = 1\} \quad (n \in \mathbb{N}^*)$$

is a subgroup of the group $(\mathbb{C}^*, \cdot)$, called the *group of n^{th} roots of unity*. Its elements are the following:

$$\varepsilon_k = \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n}, \quad k \in \{0, \dots, n-1\}.$$

- (f) Consider the general linear group $(GL_n(\mathbb{R}), \cdot)$ of rank n , where $GL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) \neq 0\}$ ($n \in \mathbb{N}$, $n \geq 2$) and denote

$$SL_n(\mathbb{R}) = \{A \in M_n(\mathbb{R}) \mid \det(A) = 1\}.$$

Then $SL_n(\mathbb{R})$ is a subgroup of $(GL_n(\mathbb{R}), \cdot)$, called the *special linear group of rank n*.

6. Subrings & subfields:

Definition

Let $(R, +, \cdot)$ be a ring and let $A \subseteq R$. Then A is called a *subring* of R if:

- (i) A is a stable subset of $(R, +, \cdot)$.
- (ii) $(A, +, \cdot)$ is a ring.

Theorem

Let $(R, +, \cdot)$ be a ring and let $A \subseteq R$. Then

$$A \text{ is a subring of } R \iff \begin{cases} A \neq \emptyset \ (0 \in A) \\ \forall x, y \in A, x - y \in A \\ \forall x, y \in A, x \cdot y \in A. \end{cases}$$

Definition

Let $(K, +, \cdot)$ be a field and let $A \subseteq K$. Then A is called a *subfield* of K if:

- (i) A is a stable subset of $(K, +, \cdot)$.
- (ii) $(A, +, \cdot)$ is a field.

Theorem

Let $(K, +, \cdot)$ be a field and let $A \subseteq K$. Then

$$A \text{ is a subfield of } K \iff \begin{cases} |A| \geq 2 \ (0, 1 \in A) \\ \forall x, y \in A, x - y \in A \\ \forall x, y \in A \text{ with } y \neq 0, x \cdot y^{-1} \in A. \end{cases}$$

- examples:

(a) Every non-trivial ring $(R, +, \cdot)$ has two subrings, namely $\{0\}$ and R , called the *trivial subrings*.

(b) $\mathbb{Z}$ is a subring of $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$ and $(\mathbb{C}, +, \cdot)$.

(c) $\mathbb{Q}$ is a subfield of $(\mathbb{R}, +, \cdot)$ and $(\mathbb{C}, +, \cdot)$, while $\mathbb{R}$ is a subfield of $(\mathbb{C}, +, \cdot)$.

(d) The set

$$n\mathbb{Z} = \{nx \mid x \in \mathbb{Z}\}$$

is a subring of $(\mathbb{Z}, +, \cdot)$ for every $n \in \mathbb{N}$. Note that $n\mathbb{Z}$ does not have identity for $n \geq 2$.

(e) The set

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z}\}$$

is a subring of the field $(\mathbb{C}, +, \cdot)$, but not a subfield. It is called the *ring of Gauss integers*.

7. Homomorphisms:

Definition

Let $(G, \cdot)$, $(G', \cdot)$ be groups and $f : G \rightarrow G'$. Then f is called a:

- ① *group homomorphism* if $f(x \cdot y) = f(x) \cdot f(y)$, $\forall x, y \in G$.
- ② *group isomorphism* if it is a bijective group homomorphism.

Theorem

Let $f : G \rightarrow G'$ be a group homomorphism. Then:

- (i) $f(1) = 1'$.
- (ii) $(f(x))^{-1} = f(x^{-1})$, $\forall x \in G$.

Definition

Let $(R, +, \cdot)$ and $(R', +, \cdot)$ be rings and $f : R \rightarrow R'$. Then f is called a:

- ① *ring homomorphism* if $\forall x, y \in R$ we have

$$\begin{aligned} f(x + y) &= f(x) + f(y), \\ f(x \cdot y) &= f(x) \cdot f(y). \end{aligned}$$

- ② *ring isomorphism* if it is a bijective ring homomorphism.

We denote by $R \simeq R'$ the fact that two rings R and R' are isomorphic.

If $f : R \rightarrow R'$ is a ring homomorphism, then the first condition from its definition tells us that f is a group homomorphism between $(R, +)$ and $(R', +)$.

Then f takes the identity element of $(R, +)$ to the identity element of $(R', +)$, that is, $f(0) = 0'$ and we also have $f(-x) = -f(x)$, $\forall x \in R$.

But in general, even if R and R' have identities, denoted by 1 and $1'$ respectively, in general it does not follow that a ring homomorphism $f : R \rightarrow R'$ has the property that $f(1) = 1'$.

Unitary ring homomorphisms

Definition

Let $(R, +, \cdot)$ and $(R', +, \cdot)$ be rings with identity elements 1 and $1'$ respectively, and let $f : R \rightarrow R'$ be a ring homomorphism. Then f is called *unitary* if $f(1) = 1'$.

Theorem

Let $(R, +, \cdot)$ and $(R', +, \cdot)$ be unitary rings with identity elements 1 and $1'$ respectively, and let $f : R \rightarrow R'$ be a ring homomorphism.

- (i) If f is surjective, then f is unitary.
- (ii) If f is a ring isomorphism, then f is unitary.
- (iii) If f is unitary and $x \in R$ has an inverse element $x^{-1} \in R$, then $f(x)$ has an inverse and

$$(f(x))^{-1} = f(x^{-1}).$$

B. VECTOR SPACES

Definition

A *vector space over K* (or a *K -vector space*) is an abelian group $(V, +)$ together with a so-called *external operation* or *scalar multiplication*

$$\cdot : K \times V \rightarrow V, \quad (k, v) \mapsto k \cdot v \quad (\text{or simply } kv),$$

satisfying the following axioms:

$$(L_1) \quad k \cdot (v_1 + v_2) = k \cdot v_1 + k \cdot v_2;$$

$$(L_2) \quad (k_1 + k_2) \cdot v = k_1 \cdot v + k_2 \cdot v;$$

$$(L_3) \quad (k_1 \cdot k_2) \cdot v = k_1 \cdot (k_2 \cdot v);$$

$$(L_4) \quad 1 \cdot v = v,$$

for every $k, k_1, k_2 \in K$ and every $v, v_1, v_2 \in V$.

The elements of K are called *scalars* and the elements of V are called *vectors*.

Sometimes a vector space is also called a *linear space*.

We usually denote a vector space V over K by ${}_K V$ or $(V, K, +, \cdot)$.

(1) In the definition of a vector space there are present four operations (3 by our definition), two denoted by the same symbol “+” and two denoted by the same symbol “·”. Of course, they are not the same, but we use the convention to denote them identically for the sake of simplicity of writing.

(2) The axioms (L_1) and (L_2) look like some distributive laws and the axiom (L_3) looks like an associative law, but they are not, since the involved elements are not taken from the same set.

(3) We have defined a *left vector space*. It is also possible to define a *right vector space* by considering an external operation

$$\cdot : V \times K \rightarrow V, \quad (v, k) \mapsto v \cdot k,$$

satisfying some similar axioms, but on the right hand side.

- basic properties:

Let V be a vector space over K . Then $\forall k, k' \in K$ and $\forall v, v' \in V$:

$$(i) \quad k \cdot 0 = 0 \cdot v = 0.$$

$$(ii) \quad k(-v) = (-k)v = -kv.$$

$$(iii) \quad k(v - v') = kv - kv'.$$

$$(iv) \quad (k - k')v = kv - k'v.$$

in

Let V be a vector space over K and let $k \in K$ and $v \in V$. Then:

$$kv = 0 \iff k = 0 \text{ or } v = 0.$$

1. Subspaces:

Definition

Let V be a vector space over K and let $S \subseteq V$. Then S is a *subspace* of V if:

$$(i) \quad S \neq \emptyset.$$

$$(ii) \quad \forall v_1, v_2 \in S, \quad v_1 + v_2 \in S.$$

$$(iii) \quad \forall k \in K, \forall v \in S, \quad kv \in S.$$

We usually denote by $S \leq_K V$, or simply by $S \leq V$, the fact that S is a subspace of the vector space V over K .

Notice that every subspace S of a vector space V over K is a subgroup of the additive group $(V, +)$, hence S must contain 0.

- characterization of subspaces:

Theorem

Let V be a vector space over K and let $S \subseteq V$. Then

$$S \leq V \iff \begin{cases} S \neq \emptyset \quad (0 \in S) \\ \forall k_1, k_2 \in K, \forall v_1, v_2 \in S, \quad k_1 v_1 + k_2 v_2 \in S. \end{cases}$$

- examples:

(a) Every non-zero vector space V over K has two subspaces, namely $\{0\}$ and V . They are called the *trivial subspaces*.

(b) Let us show that

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 0\},$$

$$T = \{(x, y, z) \in \mathbb{R}^3 \mid x = y = z\}.$$

are subspaces of the canonical real vector space $\mathbb{R}^3$ [...].

Note that S is a plane passing through the origin. For instance, the plane

$$\{(x, y, z) \in \mathbb{R}^3 \mid x + y + z = 1\}$$

is not a subspace of $\mathbb{R}^3$ over $\mathbb{R}$.

Note that T is a line passing through the origin.

(c) More generally, the only subspaces of $\mathbb{R}^3$ are $\{(0, 0, 0)\}$, any line containing the origin, any plane containing the origin and $\mathbb{R}^3$.

(d) Let $n \in \mathbb{N}$ and let

$$K_n[X] = \{f \in K[X] \mid \text{degree}(f) \leq n\}.$$

Then $K_n[X]$ is a subspace of the polynomial vector space $K[X]$ over K . Note that the set $\{f \in K[X] \mid \text{degree}(f) = n\}$ is not a subspace of $K[X]$ over K .

(e) Let $I \subseteq \mathbb{R}$ be an interval. We have seen that

$$\mathbb{R}^I = \{f \mid f : I \rightarrow \mathbb{R}\}$$

is a real vector space, where the addition and the scalar multiplication are defined as follows: $\forall f, g : I \rightarrow \mathbb{R}, \forall k \in K$, we have $f + g : I \rightarrow \mathbb{R}, kf : I \rightarrow \mathbb{R}$, where

$$(f + g)(x) = f(x) + g(x),$$

$$(kf)(x) = kf(x), \forall x \in I.$$

The subsets

$$C(I, \mathbb{R}) = \{f \in \mathbb{R}^I \mid f \text{ continuous on } I\},$$

$$D(I, \mathbb{R}) = \{f \in \mathbb{R}^I \mid f \text{ derivable on } I\}$$

are subspaces of $\mathbb{R}^I$, because they are nonempty and we have:

$$\forall k_1, k_2 \in \mathbb{R}, \forall f, g \in C(I, \mathbb{R}), k_1 f + k_2 g \in C(I, \mathbb{R}),$$

$$\forall k_1, k_2 \in \mathbb{R}, \forall f, g \in D(I, \mathbb{R}), k_1 f + k_2 g \in D(I, \mathbb{R}).$$

2. Intersection of subspaces:

For a vector space V over K , we denote by $S(V)$ the set of all subspaces of V . Sometimes, this set is denoted by $S_K(V)$ if we like to emphasize the field K .

Theorem

Let V be a vector space over K and let $(S_i)_{i \in I}$ be a family of subspaces of V . Then $\bigcap_{i \in I} S_i \in S(V)$.

3. Generated subspaces:

Definition

Let V be a vector space and let $X \subseteq V$. Then we denote

$$\langle X \rangle = \bigcap \{S \leq V \mid X \subseteq S\}$$

and we call it the *subspace generated by X* or the *subspace spanned by X* . Here X is called the *generating set* of $\langle X \rangle$.

If $X = \{v_1, \dots, v_n\}$, we denote $\langle v_1, \dots, v_n \rangle = \langle \{v_1, \dots, v_n\} \rangle$.

(1) $\langle X \rangle$ is the "smallest" (with respect to inclusion) subspace of V containing X .

(2) $\langle \emptyset \rangle = \{0\}$.

(3) If $S \leq V$, then $\langle S \rangle = S$.

Definition

A vector space V over K is called *finitely generated* if $\exists v_1, \dots, v_n \in V$ ($n \in \mathbb{N}$) such that

$$V = \langle v_1, \dots, v_n \rangle.$$

Then the set $\{v_1, \dots, v_n\}$ is called a *system of generators* for V .

Definition

Let V be a vector space over K and $v_1, \dots, v_n \in V$ ($n \in \mathbb{N}$). A finite sum of the form

$$k_1 v_1 + \dots + k_n v_n,$$

where $k_i \in K$ ($i = 1, \dots, n$), is called a (finite) *linear combination* of the vectors $v_1, \dots, v_n$.

- characterization of the generated subspaces:

Theorem

Let V be a vector space over K and let $\emptyset \neq X \subseteq V$. Then

$$\langle X \rangle = \{k_1 v_1 + \dots + k_n v_n \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^*\},$$

that is, the set of all finite linear combinations of vectors of X .

- examples:

(a) Consider the canonical real vector space $\mathbb{R}^3$. Then

$$\begin{aligned} & \langle (1, 0, 0), (0, 1, 0), (0, 0, 1) \rangle \\ &= \{k_1(1, 0, 0) + k_2(0, 1, 0) + k_3(0, 0, 1) \mid k_1, k_2, k_3 \in \mathbb{R}\} \\ &= \{(k_1, 0, 0) + (0, k_2, 0) + (0, 0, k_3) \mid k_1, k_2, k_3 \in \mathbb{R}\} \\ &= \{(k_1, k_2, k_3) \mid k_1, k_2, k_3 \in \mathbb{R}\} = \mathbb{R}^3. \end{aligned}$$

Hence $\mathbb{R}^3$ is generated by the three vectors $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, and thus it is finitely generated.

(b) Consider the canonical vector space $\mathbb{Z}_2^3$ over $\mathbb{Z}_2$. Similarly as above, we have:

$$\langle (\hat{1}, \hat{0}, \hat{0}), (\hat{0}, \hat{1}, \hat{0}) \rangle = \{(k_1, k_2, \hat{0}) \mid k_1, k_2 \in \mathbb{Z}_2\} \neq \mathbb{Z}_2^3.$$

Hence $\mathbb{Z}_2^3$ is not generated by the two vectors $(\hat{1}, \hat{0}, \hat{0})$ and $(\hat{0}, \hat{1}, \hat{0})$. But it is generated by $(\hat{1}, \hat{0}, \hat{0})$, $(\hat{0}, \hat{1}, \hat{0})$ and $(\hat{0}, \hat{0}, \hat{1})$, hence it is finitely generated.

(c) Consider the subspace

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid x - y - z = 0\}$$

of the canonical real vector space $\mathbb{R}^3$. Let us write it as a generated subspace. Expressing $x = y + z$, we have:

$$\begin{aligned} S &= \{(y + z, y, z) \mid y, z \in \mathbb{R}\} = \{(y, y, 0) + (z, 0, z) \mid y, z \in \mathbb{R}\} \\ &= \{y(1, 1, 0) + z(1, 0, 1) \mid y, z \in \mathbb{R}\} = \langle (1, 1, 0), (1, 0, 1) \rangle. \end{aligned}$$

Alternatively, one may express y or z by using the other two components and get other writings of S as a generated subspace, namely $S = \langle (1, 1, 0), (0, -1, 1) \rangle = \langle (1, 0, 1), (0, 1, -1) \rangle$. We see that S is finitely generated.

4. Sum of subspaces:

Definition

Let V be a vector space over K and let $S, T \leq V$. We define the *sum* of the subspaces S and T as the set

$$S + T = \{s + t \mid s \in S, t \in T\}.$$

Theorem

Let V be a vector space over K and $S, T \leq V$. Then $S + T = \langle S \cup T \rangle$, hence $S + T \leq V$.

- direct sum of subspaces:

Definition

Let V be a vector space over K and let $S, T \leq V$. If $S \cap T = \{0\}$, then $S + T$ is denoted by $S \oplus T$ and is called the *direct sum* of the subspaces S and T .

Theorem

Let V be a vector space over K and let $S, T \leq V$. Then

$$V = S \oplus T \iff \forall v \in V, \exists! s \in S, t \in T : v = s + t.$$

5. Linear maps:

Definition

Let V and V' be vector spaces over the same field K . A function $f : V \rightarrow V'$ is called:

(1) *(K-)linear map* (or *(vector space) homomorphism* or *linear transformation*) if

$$\begin{aligned} f(v_1 + v_2) &= f(v_1) + f(v_2), \quad \forall v_1, v_2 \in V, \\ f(kv) &= kf(v), \quad \forall k \in K, \forall v \in V. \end{aligned}$$

(2) *isomorphism* if it is a bijective K -linear map.

(3) *endomorphism* if it is a K -linear map and $V = V'$.

(4) *automorphism* if it is a bijective K -linear map and $V = V'$.

- properties of linear maps:

If $f : V \rightarrow V'$ is a K -linear map, then the first condition from its definition tells us that f is a group homomorphism between the groups $(V, +)$ and $(V', +)$. Then we have $f(0) = 0'$ and $f(-v) = -f(v)$, $\forall v \in V$.

We denote by $V \simeq V'$ the fact that two vector spaces V and V' are isomorphic. We also denote

$$\begin{aligned} \text{Hom}_K(V, V') &= \{f : V \rightarrow V' \mid f \text{ is } K\text{-linear}\}, \\ \text{End}_K(V) &= \{f : V \rightarrow V \mid f \text{ is } K\text{-linear}\}, \\ \text{Aut}_K(V) &= \{f : V \rightarrow V \mid f \text{ is bijective } K\text{-linear}\}. \end{aligned}$$

Theorem

Let V and V' be vector spaces over K and $f : V \rightarrow V'$. Then f is a K -linear map $\iff f(k_1v_1 + k_2v_2) = k_1f(v_1) + k_2f(v_2)$, $\forall k_1, k_2 \in K, \forall v_1, v_2 \in V$.

Theorem

(i) Let $f : V \rightarrow V'$ be an isomorphism of vector spaces over K . Then $f^{-1} : V' \rightarrow V$ is again an isomorphism of vector spaces over K .
(ii) Let $f : V \rightarrow V'$ and $g : V' \rightarrow V''$ be K -linear maps. Then $g \circ f : V \rightarrow V''$ is a K -linear map.

- Kernel and image of a linear map:

Definition

Let $f : V \rightarrow V'$ be a K -linear map. Then the set

$$\text{Ker } f = \{v \in V \mid f(v) = 0'\}$$

is called the *kernel* (or the *null space*) of the K -linear map f and the set

$$\text{Im } f = \{f(v) \mid v \in V\}$$

is called the *image* (or the *range space*) of the K -linear map f .

Theorem

Let $f : V \rightarrow V'$ be a K -linear map. Then

$$\text{Ker } f \leq V \text{ and } \text{Im } f \leq V'.$$

Theorem

Let $f : V \rightarrow V'$ be a K -linear map. Then

$$\text{Ker } f = \{0\} \iff f \text{ is injective}.$$

Theorem

Let $f : V \rightarrow V'$ be a K -linear map and let $X \subseteq V$. Then

$$f(\langle X \rangle) = \langle f(X) \rangle.$$

Proof. If $X = \emptyset$, then we have:

$$f(\langle \emptyset \rangle) = f(\{0\}) = \{f(0)\} = \{0'\} = \langle \emptyset \rangle = \langle f(\emptyset) \rangle.$$

If $X \neq \emptyset$, use

$$\langle X \rangle = \{k_1v_1 + \dots + k_nv_n \mid k_i \in K, v_i \in X, i = 1, \dots, n, n \in \mathbb{N}^*\} [\dots].$$

Theorem

Let V and V' be vector spaces over K . Consider on $\text{Hom}_K(V, V')$ the operations: $\forall f, g \in \text{Hom}_K(V, V')$ and $\forall k \in K$, $f + g, k \cdot f \in \text{Hom}_K(V, V')$, where

$$\begin{aligned} (f + g)(v) &= f(v) + g(v), \\ (kf)(v) &= kf(v) \end{aligned}$$

$\forall v \in V$. Then $\text{Hom}_K(V, V')$ is a vector space over K .

Corollary

Let V be a vector space over K . Then $\text{End}_K(V)$ is a vector space over K .

6. Linear independence:

Definition

Let V be a vector space over K . We say that the vectors $v_1, \dots, v_n \in V$ are (or the set of vectors $\{v_1, \dots, v_n\}$ is):

(1) *linearly independent* in V if for every $k_1, \dots, k_n \in K$,

$$k_1 v_1 + \dots + k_n v_n = 0 \implies k_1 = \dots = k_n = 0.$$

(2) *linearly dependent* in V if they are not linearly independent, that is, $\exists k_1, \dots, k_n \in K$ not all zero such that

$$k_1 v_1 + \dots + k_n v_n = 0.$$

(1) A set consisting of a single vector v is linearly dependent $\iff v = 0$.

(2) As an immediate consequence of the definition, we notice that if V is a vector space over K and $X, Y \subseteq V$ such that $X \subseteq Y$, then:

(i) If Y is linearly independent, then X is linearly independent.

(ii) If X is linearly dependent, then Y is linearly dependent. Thus, every set of vectors containing the zero vector is linearly dependent.

(3) More generally, an infinite set of vectors of V is called *linearly independent* if any finite subset is linearly independent, and *linearly dependent* if there exists a finite subset which is linearly dependent.

- linear dependence:

Let V be a vector space over K . Then the vectors $v_1, \dots, v_n \in V$ are linearly dependent if and only if one of the vectors is a linear combination of the others, that is, $\exists j \in \{1, \dots, n\}$ such that

$$v_j = \sum_{\substack{i=1 \\ i \neq j}}^n \alpha_i v_i$$

for some $\alpha_i \in K$, where $i \in \{1, \dots, n\}$ and $i \neq j$.

- linear dependence in K^n :

Theorem

Let $n \in \mathbb{N}$, $n \geq 2$.

(i) Two vectors in the canonical vector space K^n are linearly dependent $\iff$ their components are respectively proportional.

(ii) n vectors in the canonical vector space K^n are linearly dependent $\iff$ the determinant consisting of their components is zero.

7. Basis:

Definition

Let V be a vector space over K . A list of vectors $B = (v_1, \dots, v_n) \in V^n$ is called a *basis* of V if:

(i) B is linearly independent in V ;

(ii) B is a system of generators for V , that is, $\langle B \rangle = V$.

- existence of basis and proof:

Theorem

Every vector space V over K has a basis.

Proof. If $V = \{0\}$, then it has the basis $\emptyset$.

Now let $V = \langle B \rangle \neq \{0\}$, where $B = (v_1, \dots, v_n)$.

• If B is linearly independent, then B is a basis and we are done.

Suppose that the list B is linearly dependent. Then

$\exists j_1 \in \{1, \dots, n\}$ such that $v_{j_1} = \sum_{i \neq j_1}^n k_i v_i$ for some $k_i \in K$. It

follows that $V = \langle B \setminus \{v_{j_1}\} \rangle$, because every vector of V can be written as a linear combination of the vectors of $B \setminus \{v_{j_1}\}$.

• If $B \setminus \{v_{j_1}\}$ is linearly independent, it is a basis and we are done.

Otherwise, $\exists j_2 \in \{1, \dots, n\} \setminus \{j_1\}$ such that $v_{j_2} = \sum_{i \neq j_1, j_2}^n k'_i v_i$ for some $k'_i \in K$. Then $V = \langle B \setminus \{v_{j_1}, v_{j_2}\} \rangle$, because every vector of V can be written as a linear combination of the vectors of $B \setminus \{v_{j_1}, v_{j_2}\}$.

• If $B \setminus \{v_{j_1}, v_{j_2}\}$ is linearly independent, then it is a basis and we are done. Otherwise, we continue the procedure. If all the previous intermediate subsets are linearly dependent, we get to

$$V = \langle B \setminus \{v_{j_1}, \dots, v_{j_{n-1}}\} \rangle = \langle v_{j_n} \rangle.$$

• If v_{j_n} were linearly dependent, then $v_{j_n} = 0$, hence we have $V = \langle v_{j_n} \rangle = \{0\}$, contradiction. Hence v_{j_n} is linearly independent and thus forms a single element basis of V . $\square$

We shall see that a vector space may have more than one basis.

- coordinates of a vector in a basis:

Definition

Let V be a vector space over K , $B = (v_1, \dots, v_n)$ a basis of V and $v \in V$. Then the scalars $k_1, \dots, k_n \in K$ appearing in the unique writing of v as a linear combination

$$v = k_1 v_1 + \dots + k_n v_n$$

of the vectors of B are called the *coordinates of v in the basis B* .

- bases and linear maps:

Theorem

Let $f : V \rightarrow V'$ be a K -linear map and let $B = (v_1, \dots, v_n)$ be a basis of V . Then f is determined by its values on the vectors of the basis B .

Proof. [...]

Corollary

Let $f, g : V \rightarrow V'$ be K -linear maps and let $B = (v_1, \dots, v_n)$ be a basis of V . If $f(v_i) = g(v_i)$, $\forall i \in \{1, \dots, n\}$, then $f = g$.

Theorem

Let $f : V \rightarrow V'$ be a K -linear map, and let $X = (v_1, \dots, v_n)$ be a list of vectors in V .

- (i) If f is injective and X is linearly independent in V , then $f(X)$ is linearly independent in V' .
- (ii) If f is surjective and X is a system of generators for V , then $f(X)$ is a system of generators for V' .
- (iii) If f is bijective and X is a basis of V , then $f(X)$ is a basis of V' .

8. Dimension:

Definition

Let V be a vector space over K . Then the number of elements of any of its bases is called the *dimension of V* and is denoted by $\dim_K V$ or simply by $\dim V$.

If $V = \{0\}$, then V has the basis $\emptyset$ and $\dim V = 0$.

Theorem

Any two bases of a vector space have the same number of elements.

Proof. [...]

Theorem (Steinitz Theorem, Exchange Theorem)

Let V be a vector space over K , $X = (x_1, \dots, x_m)$ a linearly independent list of vectors of V and $Y = (y_1, \dots, y_n)$ a system of generators of V . Then:

- (i) $m \leq n$.
- (ii) m vectors of Y can be replaced by the vectors of X obtaining again a system of generators for V .

- characterization of dimension:

Theorem

Let V be a vector space over K . The following are equivalent:

- (i) $\dim V = n$.
- (ii) The maximum no. of linearly independent vectors in V is n .
- (iii) The minimum no. of generators for V is n .

Proof. (i) $\implies$ (ii) Assume that $\dim V = n$. Let $B = (v_1, \dots, v_n)$ be a basis of V . Then B is linearly independent in V . Since B is a system of generators for V , any linearly independent list in V must have at most n elements by Steinitz Theorem.

(ii) $\implies$ (i) Assume (ii). Let $B = (v_1, \dots, v_m)$ be a basis of V and let $(u_1, \dots, u_n)$ be a linearly independent list in V . Since B is linearly independent, we have $m \leq n$ by hypothesis. Since B is a system of generators for V , we have $n \leq m$ by Steinitz Theorem. Hence $m = n$ and consequently $\dim V = n$.

Theorem

Any linearly independent list of vectors in a vector space can be completed to a basis of the vector space.

Corollary

Let V be a vector space over K and $S \leq V$. Then:

- (i) Any basis of S is a part of a basis of V .
- (ii) $\dim S \leq \dim V$.
- (iii) $\dim S = \dim V \iff S = V$.

- decomposition theorem:

Theorem

Let V be a vector space over K and let $S \leq V$. Then there exists $\bar{S} \leq V$ such that $V = S \oplus \bar{S}$. In particular,

$$\dim V = \dim S + \dim \bar{S}.$$

Proof. Let $(u_1, \dots, u_m)$ be a basis of S . Then it can be completed to a basis $B = (u_1, \dots, u_m, v_{m+1}, \dots, v_n)$ of V . We consider

$$\bar{S} = \langle v_{m+1}, \dots, v_n \rangle$$

and we prove that $V = S \oplus \bar{S}$ [...]. $\square$

- complement of a subspace:

Definition

Let V be a vector space over K and $S \leq V$. Then a subspace $\bar{S}$ of V such that

$$V = S \oplus \bar{S}$$

is called a *complement* of S in V .

Note that a subspace may have more than one complement.

Consider the subspace $S = \langle e_1, e_2 \rangle$ of the canonical real vector space $\mathbb{R}^3$, where $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$. Then clearly (e_1, e_2) is a basis of S .

It can be completed to a basis of $\mathbb{R}^3$, with the vector $e_3 = (0, 0, 1)$ or with the vector $v_3 = (1, 1, 1)$. Following the proof of the above theorem, a complement in V of the subspace $S = \langle e_1, e_2 \rangle$ is $\langle e_3 \rangle$ or $\langle v_3 \rangle$.

- First Dimension Theorem:

Definition

Let $f : V \rightarrow V'$ be a K -linear map. Then:

- (1) $\dim(\text{Ker} f)$ is called the *nullity* of f , and is denoted by $\text{null}(f)$.
- (2) $\dim(\text{Im} f)$ is called the *rank* of f , and is denoted by $\text{rank}(f)$.

Theorem (First Dimension Theorem)

Let $f : V \rightarrow V'$ be a K -linear map. Then

$$\dim V = \dim(\text{Ker} f) + \dim(\text{Im} f).$$

In other words, $\dim V = \text{null}(f) + \text{rank}(f)$.

- Second Dimension Theorem:

Theorem (Second Dimension Theorem)

Let V be a vector space over K and let S, T be subspaces of V . Then

$$\dim S + \dim T = \dim(S \cap T) + \dim(S + T).$$

Corollary

Let V be a vector space over K , and let S and T be subspaces of V such that $V = S \oplus T$. Then

$$\dim V = \dim S + \dim T.$$

- dimension theorems:

Theorem

Let V and V' be vector spaces over K . Then

$$V \simeq V' \iff \dim V = \dim V'.$$

Proof. $\implies$ If $f : V \rightarrow V'$ is a K -isomorphism and $B = (v_1, \dots, v_n)$ is a basis of V , then one shows that $B' = f(B) = (f(v_1), \dots, f(v_n))$ is a basis of V' . [...]

$\impliedby$ If $B = (v_1, \dots, v_n)$ and $B' = (v'_1, \dots, v'_n)$ are bases of V and V' respectively, define a function $f : V \rightarrow V'$ in the following way. For every $v = k_1 v_1 + \dots + k_n v_n \in V$ (where $k_1, \dots, k_n \in K$ are uniquely determined), define $f(v) = k_1 v'_1 + \dots + k_n v'_n$. One proves that f is a K -isomorphism. [...]